

Introduction to Python Programming

Lec - 6

Getting Input from the User

- The built-in input function requests and obtains user input:
- First, input displays its string argument—called a prompt—to tell the user what to type and waits for the user to respond. We typed Paul (without quotes) and pressed Enter. We use bold text to distinguish the user's input from the prompt text that input displays.
- Function input then returns (that is, gives back) those characters as a string that the program can use. Here we assigned that string to the variable name.

```
In [1]: name = input("What's your name? ")  
What's your name? Paul
```

```
In [2]: name  
Out[2]: 'Paul'
```

```
In [3]: print(name)  
Paul
```

Function input Always Returns a String

```
In [7]: value1 = input('Enter first number: ')
Enter first number: 7

In [8]: value2 = input('Enter second number: ')
Enter second number: 3

In [9]: value1 + value2
Out[9]: '73'
```

- Rather than adding the integers 7 and 3 to produce 10, Python “adds” the string values '7' and '3', producing the string '73'. This is known as string concatenation. It creates a new string containing the left operand’s value followed by the right operand’s value.

Getting an Integer from the User

If you need an integer, convert the string to an integer using the built-in **int** function:

```
In [10]: value = input('Enter an integer: ')\nEnter an integer: 7
```

```
In [11]: value = int(value)
```

```
In [12]: value\nOut[12]: 7
```

We could have combined the code in snippets [10] and [11]:

```
In [13]: another_value = int(input('Enter another integer: '))\nEnter another integer: 13
```

```
In [14]: another_value\nOut[14]: 13
```

Getting an Integer from the User

- An alphanumeric string cannot be converted into a numeric data type.
- Forcing a conversion results in `ValueError`.

```
In [16]: bad_value = int(input('Enter another integer: '))
Enter another integer: hello
-----
ValueError                                Traceback (most recent call last)
<ipython-input-16-cd36e6cf8911> in <module>()
----> 1 bad_value = int(input('Enter another integer: '))

ValueError: invalid literal for int() with base 10: 'hello'
```

Decision Making: The if Statement and Comparison Operators

- A condition is a Boolean expression with the value True or False.

```
In [1]: 7 > 4  
Out[1]: True
```

```
In [2]: 7 < 4  
Out[2]: False
```

- True and False are keywords—words that Python reserves for its language features. Using a keyword as an identifier causes a `SyntaxError`. True and False are each capitalized.

Decision Making: The if Statement and Comparison Operators

- Operators $>$, $=$ and $<=$ all have the same precedence.
Operators $==$ and $!=$ both have the same precedence, which is lower than that of $>$, $=$ and $<=$.
A syntax error occurs when any of the operators $==$, $!=$, $>=$ and $<=$ contains spaces between its pair of symbols

Algebraic operator	Python operator	Sample condition	Meaning
$>$	<code>></code>	<code>x > y</code>	x is greater than y
$<$	<code><</code>	<code>x < y</code>	x is less than y
$\geq$	<code>>=</code>	<code>x >= y</code>	x is greater than or equal to y
$\leq$	<code><=</code>	<code>x <= y</code>	x is less than or equal to y
$=$	<code>==</code>	<code>x == y</code>	x is equal to y
$\neq$	<code>!=</code>	<code>x != y</code>	x is not equal to y

```
In [3]: 7 > = 4
File "<ipython-input-3-5c6e2897f3b3>", line 1
    7 > = 4
        ^
SyntaxError: invalid syntax
```

If, Else, Elif and the nested Ifs

- Write a program(WAP) to check if entered 0.
- WAP to check if number is 0 or non-zero
- Write a program to check if a character is vowel or not.
- WAP to check if you are eligible to drive or not. If age is ≥ 18 and has a license it must print "You can drive", otherwise "You need a license". If age is < 18 , it must print "You are too young to drive".
- WAP to find out which year of Btech you are in. The program should take semester as an input and then print the year you are in. Ensure that if invalid semester number is entered it must print "Invalid Sem".
- Evaluate the expressions
$$2 ** 3 ** 4$$
$$1 + 2 + 3 * 4 / / (2 - 4)$$
$$5 + 2 ** (3 - (-1)) * 8 / 4 ** (1/2)$$